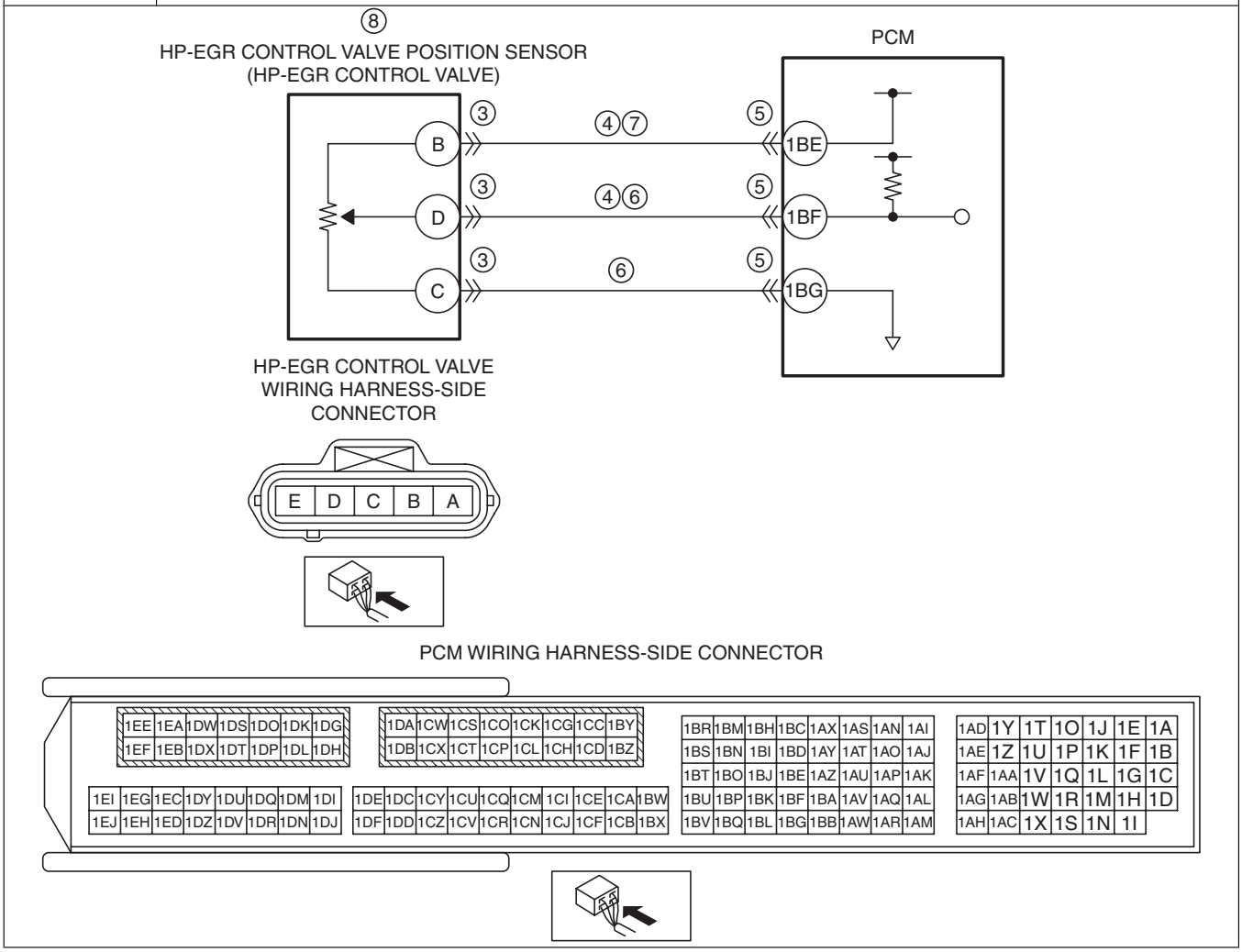
DTC P0405:00 [SKYACTIV-D 1.5]

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DTC P0405:00	HP-EGR control valve position sensor circuit low input
DETECTION CONDITION	• If the input voltage at the PCM terminal 1BF is **less than 0.25 V** for **2 s**, the PCM determines that the HP-EGR control valve position sensor circuit is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is an intermittent monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• HP-EGR control valve connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — HP-EGR control valve terminal B—PCM terminal 1BE — HP-EGR control valve terminal D—PCM terminal 1BF • PCM connector or terminals malfunction • HP-EGR control valve position sensor signal circuit and ground circuit are shorted to each other • Open circuit in wiring harness between HP-EGR control valve terminal B and PCM terminal 1BE • HP-EGR control valve position sensor malfunction • PCM malfunction

DTC P0405:00	HP-EGR control valve position sensor circuit low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT HP-EGR CONTROL VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the HP-EGR control valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT HP-EGR CONTROL VALVE POSITION SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the HP-EGR control valve connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — HP-EGR control valve terminal B — HP-EGR control valve terminal D • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — HP-EGR control valve terminal B—PCM terminal 1BE — HP-EGR control valve terminal D—PCM terminal 1BF **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 9.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT HP-EGR CONTROL VALVE POSITION SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the HP-EGR control valve and PCM connectors are disconnected. • Inspect for continuity between HP-EGR control valve terminals D and C (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • HP-EGR control valve terminal D—PCM terminal 1BF • HP-EGR control valve terminal C—PCM terminal 1BG **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 9.
		No	Go to the next step.
7	**INSPECT HP-EGR CONTROL VALVE POSITION SENSOR POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Verify that the HP-EGR control valve and PCM connectors are disconnected. • Inspect for continuity between HP-EGR control valve terminal B (wiring harness-side) and PCM terminal 1BE (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between HP-EGR control valve terminal B and PCM terminal 1BE. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT HP-EGR CONTROL VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the HP-EGR control valve position sensor. (See HP-EGR CONTROL VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction**?**	Yes	Replace the HP-EGR control valve, then go to the next step. (See HP-EGR CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.